

Clinical Trial Report

Immunexis

A Novel Immunotherapy Agent

Makana Pharma

Protocol Number: MKP-IMX-2024-001

Study Phase: Phase III

Date: January 15, 2024

Executive Summary

This report presents the results of a Phase III clinical trial evaluating the safety and efficacy of Immunexis, a novel immunotherapy agent developed by Makana Pharma for the treatment of advanced melanoma.

Study Design: Randomized, double-blind, placebo-controlled trial with 450 patients across 25 sites in North America and Europe.

Primary Endpoint: Overall survival (OS) at 24 months.

Key Findings:

• Immunexis demonstrated a statistically significant improvement in overall survival compared to placebo (HR 0.68, 95% CI: 0.55-0.84, $p < 0.001$)

• The median OS was 18.2 months in the Immunexis group vs 12.1 months in the placebo group

• Treatment-related adverse events were consistent with the known safety profile

1. Study Objectives

Primary Objective:

To evaluate the efficacy of Immunexis compared to placebo in prolonging overall survival in patients with advanced melanoma who have progressed on or after prior therapy.

Secondary Objectives:

- To assess progression-free survival (PFS)
- To evaluate objective response rate (ORR)
- To characterize the safety and tolerability profile
- To assess quality of life using standardized instruments
- To evaluate pharmacokinetics and pharmacodynamics

The study was designed with 90% power to detect a hazard ratio of 0.70 for overall survival, assuming a two-sided alpha of 0.05.

2. Patient Population

Inclusion Criteria:

- Histologically confirmed advanced melanoma (unresectable Stage III or Stage IV)
- Disease progression on or after at least one prior systemic therapy
- ECOG performance status 0-1
- Adequate organ function
- Age ≥ 18 years

Exclusion Criteria:

- Active brain metastases or leptomeningeal disease
- Active autoimmune disease requiring systemic treatment
- Prior therapy with anti-PD-1, anti-PD-L1, or anti-CTLA-4 antibodies
- Significant cardiovascular disease

A total of 450 patients were randomized in a 2:1 ratio to receive either Immunexis 240mg IV every 2 weeks (n=300) or placebo (n=150). Baseline characteristics were well balanced between treatment arms.

3. Efficacy Results

3.1 Overall Survival

The primary endpoint of overall survival was met with statistical significance. At the pre-specified interim analysis, the independent data monitoring committee recommended early termination of the trial due to clear demonstration of efficacy.

Results at 24 months:

• Immunexis arm: 42% (95% CI: 37-47%)

• Placebo arm: 28% (95% CI: 21-35%)

• Hazard ratio: 0.68 (95% CI: 0.55-0.84, $p < 0.001$)

The benefit of Immunexis was observed across all pre-specified subgroups, including age, gender, baseline disease burden, and prior therapy.

3.2 Progression-Free Survival

Secondary endpoint analysis showed:

• Median PFS: 8.4 months (Immunexis) vs 4.2 months (placebo)

• HR: 0.58 (95% CI: 0.48-0.70, $p < 0.001$)

4. Safety and Tolerability

Treatment-Emergent Adverse Events (TEAEs):

The safety profile of Immunexis was consistent with the known class effects of immunotherapy

agents. The most common TEAEs (≥20%) in the Immunexis arm were:

• Fatigue: 45%

• Nausea: 28%

• Diarrhea: 24%

• Rash: 22%

• Pruritus: 21%

Immune-Related Adverse Events (irAEs):

Grade 3-4 irAEs occurred in 18% of patients in the Immunexis arm vs 3% in placebo:

• Colitis: 4%

• Hepatitis: 3%

• Pneumonitis: 2%

• Hypophysitis: 1%

All irAEs were managed with established protocols including corticosteroids and temporary treatment interruption. There were 2 treatment-related deaths in the Immunexis arm (pneumonitis and myocarditis).

Treatment discontinuation due to adverse events occurred in 12% of Immunexis patients vs 5% of placebo patients.

5. Conclusions

This Phase III clinical trial demonstrates that Immunexis significantly improves overall survival and progression-free survival in patients with advanced melanoma who have progressed on prior therapy. The magnitude of benefit (32% reduction in risk of death) represents a clinically meaningful improvement.

The safety profile is manageable and consistent with other immunotherapy agents in this class. The incidence of immune-related adverse events can be effectively managed with established protocols.

Key Strengths:

- Robust study design with appropriate statistical power
- Well-balanced baseline characteristics
- Comprehensive safety monitoring
- Consistent benefit across subgroups

These results support the use of Immunexis as a new treatment option for patients with advanced melanoma and form the basis for regulatory submission to the FDA and EMA.

Next Steps:

- Prepare regulatory submission package
- Continue long-term follow-up for durability of response
- Explore combination strategies in earlier disease stages